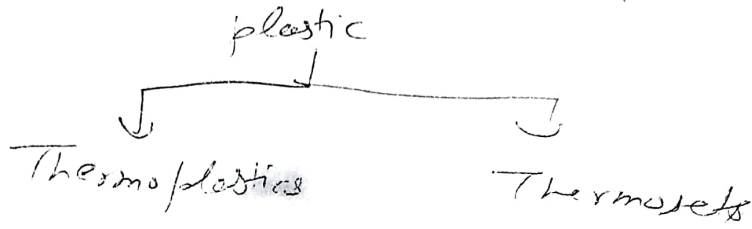
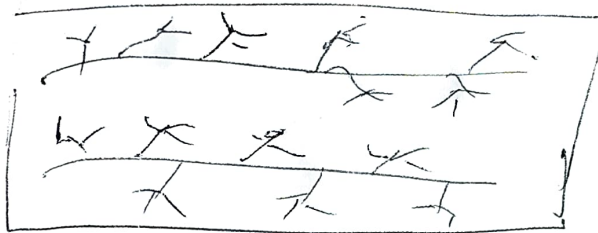


1 Plastic → word comes from the greek word *plastikos* meaning "able to be shaped and molded". (1)



Thermoplastics → A material that cures reversibly and softens when heated above the glass transition temp. or melting point and become hard after cooling is called thermoplastic.

→ It has linear or branched molecular structure.



Properties of thermoplastics

- ① High strength & toughness.
- ② Better hardness
- ③ Chemical resistance.
- ④ Durability
- ⑤ Self lubrication, Transparency & Water proof.

Can be made transparent or coloured,
 light in wt because of low density
 silent in operations
 Easy workability
 Low melting point
 High resistance abrasion
 Good surface finish
 Good thermal & electrical insulation
 Good strength and rigidity
 Good resistance to most of the chemicals
 Good dimension stability
 Low fabrication cost

Type of Thermoplastic material

- 1) Acrylonitrile butadiene styrene (ABS)
- 2) Acrylics

- ③ Polycarbonates (PC); Polyethylene PE
- ④ Polypropylene (PP) ②
- ⑤ Polyvinyl chloride (PVC); Polyester
- ⑥ liquid crystal polymers (LCP); Polystyrene
- ⑦ Vinyls.

Thermoset → The property of material permanently hard and rigid after cooling when heated above the melting temp is called thermosets.

→ The solidification process of plastic is known as Curing.

→ The transformation from the liquid state to the solid state is irreversible process.

→ It can not be recycled.

→ During curing, the small molecules are chemically linked together to form complex interconnected network structure. The cross-linking prevents the slippage of individual chain. Therefore the mechanical properties (tensile strength, compressive strength and hardness) are not temp. dependent, as compared to thermoplastic. Hence, thermosets are generally stiffer than the thermoplastics.

Types Alkyds, Alkylics, Amine, Cellulose, epoxy, phenolic (PIE), polyester & Vinyl ester etc.

Applications high temp application like - electrical equipment, motor brush holder, circuit breaker handle and knobs, kitchen utensils.

Processing technique for plastics

- Heating, forming and cooling in continuous or repeated cycle.
- Thermoplastic can be processed by heating up to glass transition temp and formed into desired shape with the application of pressure Ex Thermo forming, Compression molding & blow film extrusion.
- The processing of thermosets can be completed in two steps. First it can be melted and then poured in to mold to make a desired shape Ex Casting, extrusion & injection molding.

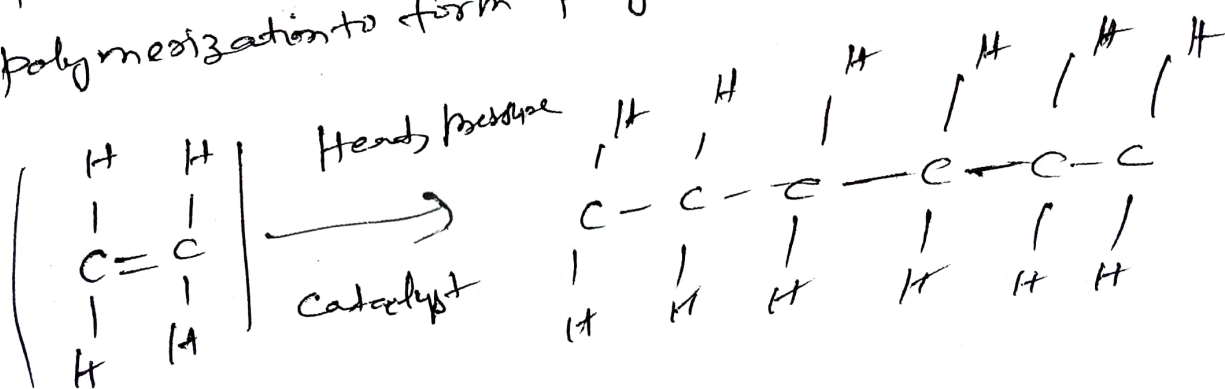
→ A single repeating unit in a molecular chain is called monomers.

Polyaddition is the monomers bond to each other without the loss of any other atoms. Ex ⁽⁹⁾ Alkane.

PolyCondensation Two different monomers combine to form a single molecules with loss of atoms. Ex formation of polyester lost hydrogen molecules.

The monomers are organic carbon based molecules. Covalent bond hold the atoms in the molecules together and secondary bond hold group of chain together.

Ex polyethylene is made from ethylene monomers. The two monomers are joined together with the application of heat, pressure & catalyst which is called polymerization to form polyethylene



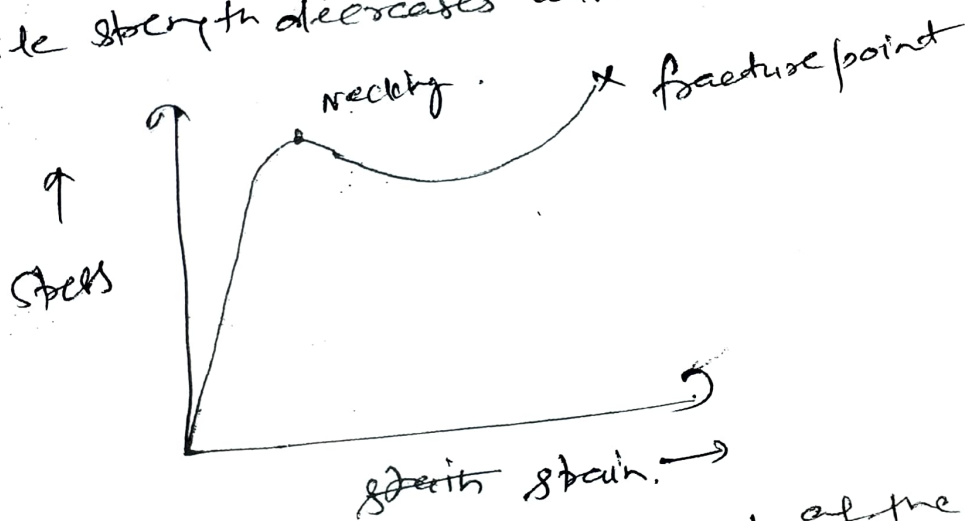
ethylene monomer unit

Mechanical properties of plastic

(5)

(a) Hardness plastic are not very hard, their hardness being comparable to that of brass and Al. Thermoplastics are harder than thermoplastic. The temp. of material substantially affects its properties. The hardness of commonly used plastic is in the range of 5-50 BHN.

(b) Stress-strain behavior The modulus of elasticity for plastic is in the range of 10 MPa to 4000 MPa. Tensile strength decreases with increase temp.



(c) Creep → The molecular weight of the polymer affects its creep behavior. As the molecular weight of the polymer increases, it becomes more creep resistant.

(a)

fatigue behavior → Fatigue failure is brittle in nature, but in case of thermoplastic, failure occurs due to initiation of crack propagation.

Casting → Plastic material is melted and it is poured in to the mold cavity. The molded part is cooled and solidifies in the mold cavity & molded part is ejected.

→ Generally air cooling is preferred.

↳ A mold release spray is used on the mold surface prior to casting.

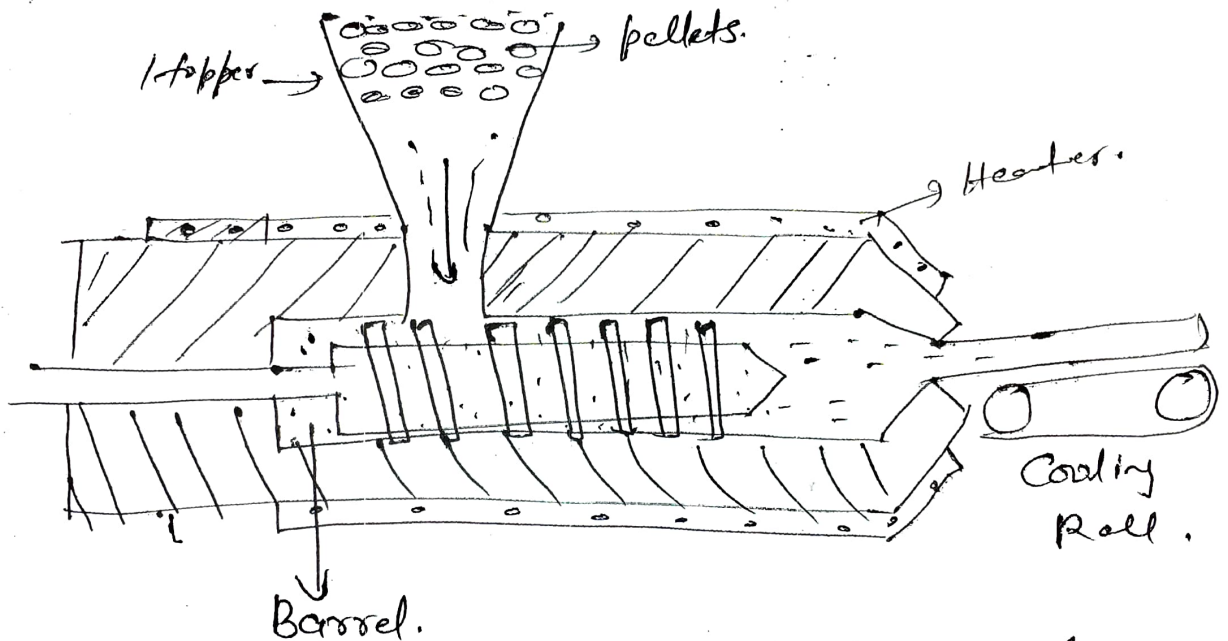
Process parameters →

- ① melting temp of plastic.
- ② pouring temp.
- ③ cooling rate.

Extrusion of plastic →

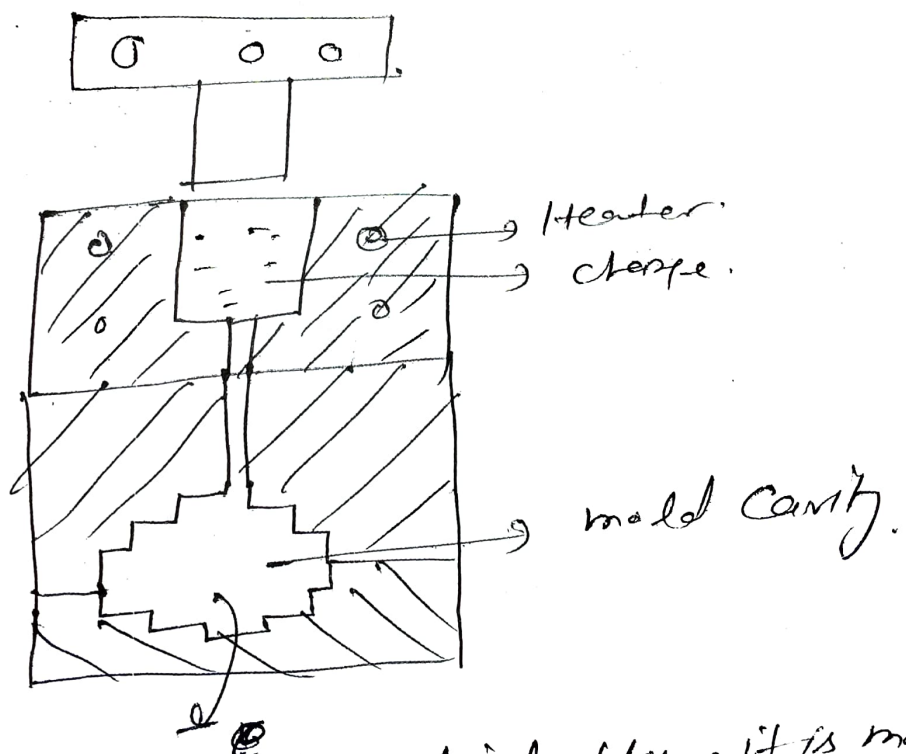
↳ It is a high volume manufacturing process. Plastic material in the form of pellets or granules is gravity fed from a hop mounted hopper in to the barrel. Additive such as colorant and ultraviolet light inhibitors can be mixed in to hopper. The plastic material enters through the feed throat and comes in to contact with the rotating screws. The screw pushes the plastic bead forward in to barrel. The barrel is heated using the heating element up to melting temperature of the plastic. The heating element are used in such a way that gradually increase the temp. of the barrel from rear to the front.

There are three possible zone in a extruding (X)
 Screws i.e feed zone, melting zone & metering zone. In the
 feed zone the plastic beads melt gradually as they are pushed
 through the barrel. The plastic material is completely
 melted in the melting zone. At the front of barrel, the molten
 plastic leaves the screws and barrel through a screen plate
 to remove any contaminants in the molten plastic. After that,
 molten plastic enters in to die. The die give the desired
 shape of plastic product. Plastic are very good thermal
 insulator & therefore it is very difficult to cool quickly. The
 plastic product is cooled by pulling through a set of cooling
 rolls.



Screw has mainly three different function namely
 feeding mechanism, uniform melting & mixing of plastic and
 finally it generate pressure to push the molten material
 through die. $L \rightarrow$ length of screw, $D \rightarrow$ Dia. of screw.
 Generally L/D ratio is used as 24:1, but more mixing.
 and out put it may increase up to 32:1.

Transfer molding Transfer molding The required amount of polymer charge is weighed & inserted into transfer pot before the molding process. The transfer pot is heated by the heating element above the melting temp of polymer charge. The liquid charge is gravity filled. The plunger is used to liquid polymer charge from the transfer pot into mold cavity under pressure. The cavity is held closed until the resin get cured. The cavity is opened and the molded part can be removed once it has hardened with the ejector pins. The Sprue and gate attached to the molded part have to be trimmed after the process has been completed.



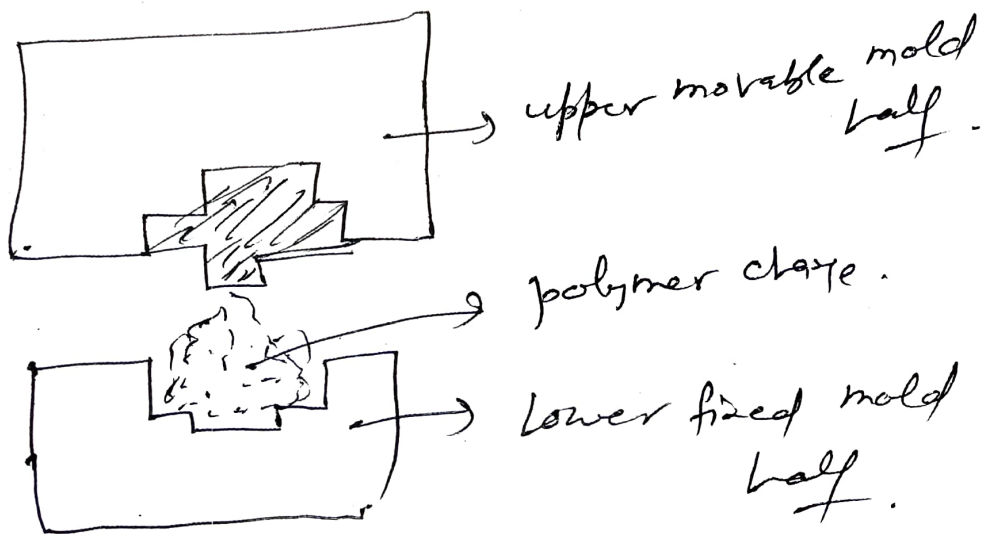
→ produce more waste material, hence it is more expensive process.

→ Generally Thermoset plastic are processed by transfer molding process.

Compression molding

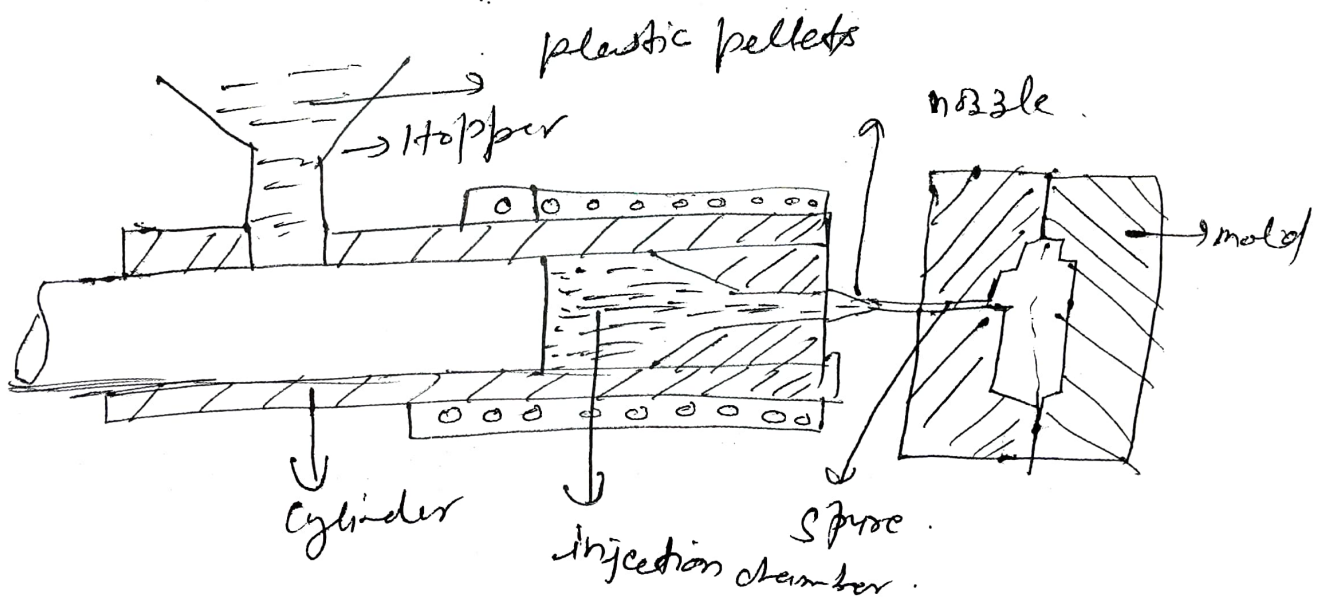
(9)

- The predetermined amount of charge of plastic material is placed into lower half of a heated mold cavity.
- The mold cavity is closed with upper movable half mold & pressure is applied to compress the material into the mold cavity. This causes the raw material to be squeezed out to take the shape of the mold cavity. The application of the heat & pressure increases the polymerization process. Hence plastic material is cured. Generally the hydraulic pressure is required in the range of 7-25 MPa to squeeze the plastic material. The temp of mold cavity is usually in the range of 130-250°C. The mold cavity is then cooled for sometimes so that molded plastic parts get solidified. The manufacturing cycle time (heating, cooling, & part ejection) may be long (about 1-6 min).



Injection Molding

- Used thin walled plastic parts for a wide variety of shapes & size.
- Plastic material in the form of powder or pellets are fed from hopper into injection chamber.
- The piston and cylinder arrangement is used to forward the material inserted from the hopper into injection chamber.
- The plastic material is heated in the injection chamber with the application of heating element.
- The molten plastic material is then injected into mold cavity through nozzle. The molded part is cooled quickly in the mold. The process cycle for injection molding is very short typically b/w 2 to 60 sec.



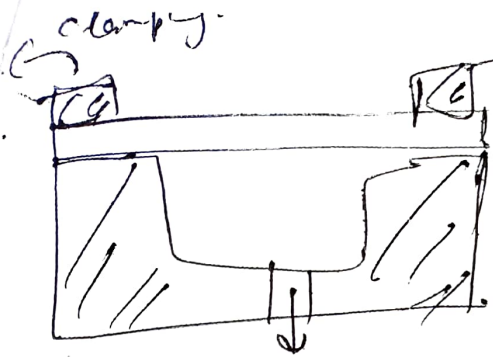
Thermoforming $\Rightarrow$ In which thermo plastic sheet are formed with the application of heat & pressure in a mold. The thermoplastic sheet is held horizontally over a mold surface and clamped with a holding device. The sheet is heated up to predetermined temperature using a heating element called heater. The thermoplastic sheet softens with the application of heat & is pressed in to or stretched over the mold surface by application of air pressure or by any other means. The softened sheet conforms to the mold shape and it is held in place until it cools. The excess material is then trimmed out from the formed part. Thin sheet (up to 1.5 mm) & thick sheet (about 3 mm) can be formed easily.

Three different type of thermo forming process

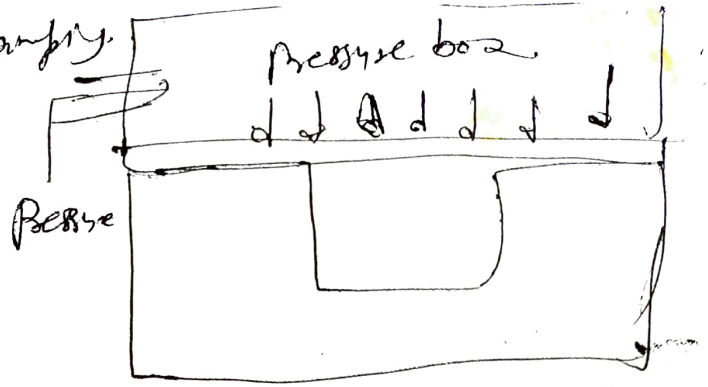
$\Rightarrow$ Vacuum forming ② pressure forming & matched die forming

① Vacuum forming The thermoplastic sheet is placed on the mold surface and fixed with the clamping unit. The sheet is heated until it is softened and thereafter vacuum need to be applied quickly.

② Pressure forming The preheated plastic sheet is placed on the mold surface & then air pressure is applied quickly above the sheet.



Vacuum



Pressure

matched die forming is also called mechanical forming.
die & punch are used to deform the softened thermoplastic sheet.